

Zachary F. Mainen

Champalimaud Research

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Personal

Citizenship: United States of America / Portugal

Date/Place of Birth: 20 February, 1969, Rockville, MD, United States of America

Married: Susana Q. Lima

Education

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| 1991-1995 | Ph.D., Neurosciences
University of California, San Diego
La Jolla, CA, USA |
| 1987-1991 | M.Sc., Psychology
Yale University
New Haven, CT, USA |
| 1987-1991 | B.A., Psychology (summa cum laude)
Yale University
New Haven, CT, USA |

Research and Professional Experience

- | | |
|--------------|---|
| 2016-present | Founder and Executive Board member
International Brain Laboratory |
| 2014-2019 | Founder and Co-Director
Champalimaud Research
Champalimaud Centre for the Unknown, Lisbon, Portugal |
| 2010-2014 | Director
Champalimaud Neuroscience Programme
Champalimaud Centre for the Unknown, Lisbon, Portugal |
| 2010-present | Senior Investigator
Champalimaud Neuroscience Programme
Champalimaud Centre for the Unknown, Lisbon, Portugal |
| 2008-2010 | Adjunct Associate Professor
Cold Spring Harbor Laboratory, Cold Spring Harbor, NY, USA |
| 2008-2010 | Visiting Scientist
Howard Hughes Medical Institute |

Janelia Farm Research Campus, Ashburn, VA, USA

2007-2011	Founder, Coordinator and Senior Group Leader Champalimaud Neuroscience Program Instituto Gulbenkian de Ciência, Oeiras, Portugal
2004-2007	Associate Professor Cold Spring Harbor Laboratory, Cold Spring Harbor, NY, USA
2002-2007	Affiliated Associate Professor Department of Neurobiology & Behavior Stony Brook University, Stony Brook, NY, USA
1999-2003	Assistant Professor Cold Spring Harbor Laboratory, Cold Spring Harbor, NY, USA
1997-1999	Postdoctoral fellow Supervisor: Karel Svoboda Cold Spring Harbor Laboratory, Cold Spring Harbor, NY, USA
1995-1997	Postdoctoral fellow Supervisor: Roberto Malinow Cold Spring Harbor Laboratory, Cold Spring Harbor, NY, USA
1992-1995	Doctoral student Supervisor: Terrence J Sejnowski The Salk Institute, La Jolla, CA, USA
1989-1991	Undergraduate research assistant Supervisor: Thomas H Brown Yale University, New Haven, CT, USA

Fellowships & Awards

2015-2020	Advanced Investigator Grant from the European Research Council (ERC)
2010	Elected Member, European Molecular Biology Organization (EMBO)
2010-2015	Advanced Investigator Grant from the European Research Council (ERC)
2005	NARSAD Lattner Foundation Investigator
2001-2005	Searle Scholar
1997-2002	Burroughs-Wellcome Career Development Award
1995-1997	NRSA Postdoctoral Fellowship
1991-1995	Howard Hughes Medical Institute Predoctoral Fellowship
1991	Phi Beta Kappa
1990	Sigma Xi

Research Support

2020	Co-PI International Brain Laboratory, Wellcome Trust, UK: £9,599,726
2018	Fundação para a Ciência e a Tecnologia, Portugal (32068/2017): €228,071.26
2018	Co-PI Fundação para a Ciência e a Tecnologia, Portugal (31331/2017): €239.569,41

2018	Co-PI Fundação para a Ciência e a Tecnologia, Portugal (28830/2017): €238,708.79
2018	Co-PI Fundação para a Ciência e a Tecnologia, Portugal (28509/2017): €239,921.28
2018	Co-PI Fundação para a Ciência e a Tecnologia, Portugal (34418/2017): €239,160.06
2017	Co-PI International Brain Laboratory, Wellcome Trust, UK: £1,998,000
2017	Simons Foundation, USA: \$702,000.00
2016	European Research Council Advanced Grant, European Union: €2,486,074.00
2014	Simons Foundation, USA: \$534,285.29
2013	Human Brain Project, European Union: €225,000
2012	Fundação para a Ciência e a Tecnologia, Portugal: €73,621
2010	Human Frontier Science Program, International: €284,853
2009	European Research Council Advanced Grant, European Union: €2,318,636.40
2007	Marie Curie Actions IEF, European Union: €92,077.94
2004	USAMRAA NETRP, USA: \$118,105
2001	David & Lucile Packard Foundation, USA: \$843,852
2001	NIH/NIDCD, USA: \$1,259,410

Memberships

2014	The American Physiological Society
1991-present	Society for Neuroscience

Advisory Boards and Review Panels

2021-present	Scientific Advisory Board Member, Max Planck Institute for Brain Research
2019-2020	Advisory Board Member, Review Commons
2018-2020	Scientific Advisory Board, COMPASS Pathways Limited
2016-present	Editorial Board Member, Current Opinion in Neurobiology
2015-present	Editorial Board Member, Neural Computation
2014-present	Editorial Board Member, Current Biology
2013-present	European Research Council Expert Panel
2013-2018	Conselho da Faculdade de Ciências da Universidade Nova de Lisboa, Portugal
2013	Advisory Board Member, Sociedade Portuguesa de Neurociências
2010-2016	Editorial Advisory Panel, Nature Communications, Nature Publishing Group
2011	Scientific Committee, Champalimaud Neuroscience Symposium
2007-2014	Associate Editor, Frontiers in Neuroscience
2005-present	Executive Board, Computational and Systems Neuroscience (Cosyne)
2005	Ad-hoc reviewer, Somatosensory and Chemosensory Study Section
2005	Ad-hoc reviewer, ZDC1 SRB-Y (R03 special emphasis panel)
2004	Ad-hoc reviewer, ZRG1 study section (Human Brain Project)

Reviewer (partial list)

Cell, e-Life, Nature, Neuron, Nature Neuroscience, Journal of Neuroscience, Journal of Neurophysiology, Science

Articles

65. Koch C, Svoboda K, Bernard A, Basso MA, Churchland AK, Fairhall AL, Groblewski PA, Lecoq JA, Mainen ZF, Mathis MW, Olsen SR, Phillips JW, Pouget A, Saxena S, Siegle JH, Zador AM (2022) Next-generation brain observatories. *Neuron*, 110.

64. Ligneul R, Mainen ZF, Ly V, Cools R (2022) Stress-sensitive inference of task controllability. *Nat Hum Behav* 6, 812.
63. Gutierrez-Castellanos N, Sarra D, Godinho BS, Mainen ZF (2022) Maturation of prefrontal input to dorsal raphe increases behavioral persistence in mice. *bioRxiv*. doi: <https://doi.org/10.1101/2022.01.01.474690>
62. Poo C, Agarwal G, Bonacchi N, Mainen ZF (2021) Spatial maps in olfactory cortex during olfactory navigation. *Nature* 601, 595-599.
61. Recanatesi S, Pereira-Obilinovic U, Murakami M, Mainen ZF, Mazucatto L (2021) Metastable attractors explain the variable timing of stable behavioral action sequences. *Neuron* 110 (1): 139.
60. Cazettes F, Murakami M, Renart A, Mainen ZF (2021) Reservoir of decision strategies in the mouse brain. *bioRxiv* doi: <https://doi.org/10.1101/2021.04.01.438090>
59. Snitz K, Honigstein D, Weissgross R, Ravia A, Mishor E, Perl O, Karagach S, Medhanie A, Harel N, Shushan S, Roth Y, Iravani B, Arshamian A, Ernst G, Okamoto M, Poo C, Bonacchi N, Mainen ZF, Monteleone E, Dinnella C, Spinelli S, Mariño-Sánchez F, Ferdenzi C, Smeets M, Touhara K, Bensafi M, Hummel T, Lundström JN, Sobel N (2021). A Novel Olfactory Self-Test Effectively Screens for COVID-19. *Commun Med* 2, 34.
58. Roy NA, Bak JH, International Brain Laboratory, Akrami A, Brody C, Pillow JW (2021) Extracting the dynamics of behavior in sensory decision-making experiments. *Neuron* 109 (4), 597-610.
57. The IBL, Aguillon-Rodriguez V, Angelaki DE, Bayer HM, Bonacchi N, Carandini M, Cazettes F, Chapuis GA, Churchland AK, Dan Y, Dewitt EE, Faulkner M, Forrest H, Haetzel LM, Hausser M, Hofer SB, Hu F, Khanal A, Krasniak CS, Laranjeira I, Mainen ZF, Meijer GT, Miska NJ, Masic-Flogel TD, Murakami M, Noel JP, Pan-Vazquez A, Sanders JI, Socha KZ, Terry R, Urai AE, Vergara HM, Wells MJ, Wilson CJ, Witten IB, Wool LE, Zador A (2021) A standardized and reproducible method to measure decision-making in mice. *eLife* 2021; 10:e63711. doi: 10.7554/eLife.63711
56. Cazettes F, Reato D, Morais JP, Renart A, Mainen ZF (2021) Phasic activation of dorsal raphe serotonergic neurons increases pupil-linked arousal. *Current Biology*, 31, 192–197.
55. Urai A, Aguillon-Rodriguez V, Laranjeira IC, Cazettes F, The IBL, Mainen ZF, Churchland AK (2021) Citric Acid as an alternative to water restrictions for high-yield mouse behavior. *eNeuro* 11 January 2021, ENEURO.0230-20.2020.
54. Ashwood ZC, Roy NA, Stone IR, International Brain Laboratory, Churchland AK, Pouget A, Pillow JW (2020) Mice alternate between discrete strategies during perceptual decision-making. *bioRxiv*, <https://doi.org/10.1101/2020.10.19.346353>
53. Iravani B, Arshamian A, Ravia A, Mishor E, Snitz K, Shushan S, Roth Y, Perl O, Honigstein D, Weissgross R, Karagach S, Ernst G, Okamoto M, Mainen ZF, Monteleone E, Dinnella C, Spinelli S, Marino-Sanchez F, Ferdenzi C, Smeets M, Touhara K, Bensafi M, Hummel T, Sobel N, Lundström JN (2020) Relationship Between Odor Intensity Estimates and COVID-19 Prevalence Prediction in a Swedish Population. *Chemical Senses* 45 (6), 449–456.
52. Mendonça AG, Drugowitsch J, Vicente M, DeWitt E, Pouget A, Mainen ZF (2020) The impact of learning on perceptual decisions and its implication for speed-accuracy tradeoffs. *Nat Comm* 11 (1): 2757.
51. Fonseca MS, Bergomi M, Mainen ZF, Shemesh N (2020) Functional MRI of large scale activity in behaving mice. *bioRxiv*, doi: <https://doi.org/10.1101/2020.04.16.044941>.

50. Vertechi P, Lottem E, Sarra D, Godinho B, Treves I, Quendera T, Oude Lohuis MN, Mainen ZF (2020) Inference-based decisions in a hidden state foraging task: differential contributions of prefrontal areas. *Neuron* 106 (1), 166.
49. International Brain Laboratory, Bonacchi N, Chapuis G, Churchland A, Harris KD, Rossant C, Sasaki M, Shen S, Steinmetz NA, Walker EY, Winter O, Wells M (2019) Data architecture and visualization for a large-scale neuroscience collaboration. *bioRxiv*, 827873
48. Drugowitsch J, Mendonça AG, Mainen ZF, Pouget A (2019) Learning optimal decisions with confidence. *PNAS* 116 (49), 24872.
47. Pardo-Vazquez JL, Castiñeiras-de Saa JR, Valente M, Damião I, Costa T, Vicente MI, Mendonça AG, Mainen ZF, Renart A (2019) The mechanistic foundation of Weber's law *Nat. Neurosci.* 22 (9).
46. Rigato J, Rennie SM, Mainen ZF (2019) The overlooked ubiquity of first-person experience in the cognitive sciences *Synthese* (doi:<https://doi.org/10.1007/s11229-019-02136-6>)
45. Igaya K, Fonseca MS, Murakami M, Mainen ZF, Dayan P (2018) An effect of serotonergic stimulation on learning rates for rewards apparent after long intertrial intervals. *Nat Commun* 9 (1), 2477.
44. Lottem E, Banerjee D, Vertechi P, Sarra D, Oude Lohuis M, Mainen ZF (2018) Activation of serotonin neurons promotes active persistence in a probabilistic foraging task. *Nat Commun* 9 (1): 1000.
43. Correia, PA, Matias, S, Mainen ZF. (2017) Stereotaxic adeno-associated virus injection and cannula implantation in the dorsal raphe nucleus of mice. *Bio-protocol* 7 (18), e2549.
42. Murakami M, Shteingart H, Loewenstein Y, Mainen ZF. (2017) Distinct sources of deterministic and stochastic components of action timing decisions in rodent frontal cortex. *Neuron* 94 (4), 908-919.e7.
41. Matias S, Lottem E, Dugué GP, Mainen ZF. (2017) Activity patterns of serotonin neurons underlying cognitive flexibility. *eLife* pii: e20552.
40. Correia PA, Lottem E, Banerjee D, Machado AS, Carey MR, Mainen ZF. (2017) Transient inhibition and long-term facilitation of locomotion by phasic optogenetic activation of serotonin neurons. *eLife* pii: e20975.
39. Grabska-Barwińska A, Barthelmé S, Beck J, Mainen ZF, Pouget A, Latham PE (2017) A probabilistic approach to demixing odors. *Nat Neurosci* 20, 98–106.
38. Kobak D, Brendel W, Constantinidis C, Feierstein CE, Kepecs A, Mainen ZF, Romo R, Qi XL, Uchida N, Machens CK. (2016) Demixed principal component analysis of neural population data. *eLife* (5), e10989.
37. Lottem E, Lörincz ML, Mainen ZF. (2016) Optogenetic activation of dorsal raphe serotonin neurons rapidly inhibits spontaneous but not odor-evoked activity in olfactory cortex. *J. Neurosci.* 36 (1), 7-18.
36. Rigato J, Murakami M, Mainen Z (2015) Spontaneous Decisions and Free Will: Empirical Results and Philosophical Considerations. *Cold Spring Harb Symp Quant Biol.*
35. Bush PC, Mainen ZF (2015) Columnar architecture improves noise robustness in a model cortical network. *PLoS One* 10 (3), e0119072.
34. Fonseca MS, Murakami M, Mainen ZF (2014) Activation of dorsal raphe serotonergic neurons promotes persistent waiting independently of reinforcing effects. *Current Biology* 25 (3), 306-15.

33. Murakami M, Vicente MI, Costa GM, Mainen ZF (2014) Neural antecedents of self-initiated actions in secondary motor cortex. *Nat Neurosci.* 11, 1574-82.
32. Lak A, Costa GM, Romberg E, Koulakov AA, Mainen ZF, Kepecs A (2014) Orbitofrontal Cortex Is Required for Optimal Waiting Based on Decision Confidence. *Neuron* 84, 190-201.
31. Dugué GP, Lörincz ML, Lottem E, Audero E, Matias S, Correia PA, Léna C, Mainen ZF. (2014) Optogenetic recruitment of dorsal raphe serotonergic neurons acutely decreases mechanosensory responsivity in behaving mice. *PLoS ONE* 9 (8), e105941.
30. Tecuapetla F, Matias S, Dugue GP, Mainen ZF, Costa RM (2014) Balanced activity in basal ganglia projection pathways is critical for contraversive movements. *Nat Commun.* 5, 4315.
29. Rennie SM, Moita MM, Mainen ZF (2013) Social cognition in the rodent: nothing to be sniffed at. *Trends Cogn Sci.* 17 (7), 306-7.
28. Zariwala HA, Kepecs A, Uchida N, Hirokawa J, Mainen ZF (2013) The limits of deliberation in a perceptual decision task. *Neuron* 78 (2), 339-51.
27. Miura K, Mainen ZF, Uchida N (2012) Odor Representations in Olfactory Cortex: Distributed Rate Coding and Decorrelated Population Activity. *Neuron* 74 (6), 1087-98.
26. Felsen G, Mainen ZF (2012) Midbrain contributions to sensorimotor decision-making. *J Neurophysiol.* 108 (1), 135-47.
25. Quirk MC, Sosulski DL, Feierstein CE, Uchida N and Mainen ZF (2009) A defined network of fast-spiking interneurons in orbitofrontal cortex: responses to behavioral contingencies and ketamine administration. *Front Syst Neurosci.* Nov09 (3), 13.
24. Ranade SP, Mainen ZF (2009) Transient firing of dorsal raphe neurons encodes diverse and specific sensory, motor, and reward events. *J. Neurophysiol.* 102, 3026-37.
23. Huber D, Petreanu L, Ghitani N, Ranade S, Hromádka T, Mainen Z, Svoboda K (2008) Sparse optical microstimulation in barrel cortex drives learned behavior in freely moving mice. *Nature* 451, 61-4.
22. Uchida N, Mainen ZF (2008) Odor concentration invariance by chemical ratio coding. *Front Sys Neurosci.* Apr 08 (1), 3.
21. Felsen G, Mainen ZF (2008) Neural substrates of sensory-guided locomotor decisions in the rat superior colliculus. *Neuron* 60, 137-48.
20. Kepecs A, Uchida N, Zariwala HA, Mainen ZF (2008) Neural correlates, computation and behavioral impact of decision confidence. *Nature* 455, 227-31.
19. Xiao Yun, Donghwi Kim, Stanacevic M., Mainen ZF. (2007) Low-power high-resolution 32-channel neural recording system. *Conf Proc IEEE Eng Med Biol Soc* , 2373-2376.
18. Sato TR, Gray NW, Mainen ZF, Svoboda K (2007) The functional microarchitecture of the mouse barrel cortex. *PLoS Biol* 5 (7), e189.

17. Kepecs AC, Uchida N, Mainen ZF (2007) Rapid and precise control of sniffing during olfactory discrimination in rats. *J Neurophys.* 98, 205-13.
16. Gurden H, Uchida N, Mainen ZF (2006) Sensory-evoked intrinsic optical signals in the olfactory bulb are coupled to glutamate release and uptake. *Neuron* 52, 335-45.
15. Feierstein CE, Quirk MC, Uchida N, Sosulski DL, Mainen ZF (2006) Representation of spatial goals in rat orbitofrontal cortex.. *Neuron* 51, 495-507.
14. Egger V, Svoboda K, Mainen ZF (2005) Dendrodendritic synaptic signals in olfactory bulb granule cells: local spine boost and global low-threshold spike. *J Neurosci.* 25, 3521-30.
13. Uchida N, Mainen ZF (2003) Speed and accuracy of olfactory discrimination in the rat. *Nature Neurosci.* 6, 1224-9.
12. Egger V, Svoboda K, Mainen ZF (2003) Mechanisms of lateral inhibition in the olfactory bulb: efficiency and modulation of spike-evoked calcium influx into granule cells. *J Neurosci.* 23, 7551-8.
11. Maravall M, Mainen ZF, Sabatini BL, Svoboda K (2000) Estimating intracellular calcium concentrations and buffering without wavelength rationing. *Biophys J.* 78, 2655-67.
10. Mainen ZF, Maletic-Savatic M, Shi SH, Hayashi Y, Malinow R, Svoboda K (1999) Two-photon imaging in brain slices. *Methods* 18, 231-9.
9. Mainen ZF, Malinow R, Svoboda K (1999) Synaptic calcium transients in single spines indicate that NMDA receptors are not saturated. *Nature* 399, 151-5.
8. Mainen ZF, Jia Z, Roder J, Malinow R (1998) Use-dependent AMPA receptor block in mice lacking GluR2 suggests postsynaptic site for LTP expression. *Nature Neurosci.* 1, 579-86.
7. Mainen ZF, Carnevale NT, Zador AM, Claiborne BJ, Brown TH (1996) Electrotonic architecture of hippocampal CA1 pyramidal neurons based on three-dimensional reconstructions. *J Neurophysiol.* 76, 1904-23.
6. Mainen ZF, Sejnowski TJ (1996) Influence of dendritic structure on firing pattern in model neocortical neurons. *Nature* 382, 363-6.
5. Malinow R, Mainen ZF (1996) Long-term potentiation in the CA1 hippocampus [technical comment]. *Science* 271, 1604-6.
4. Mainen ZF, Joerges J, Huguenard JR and Sejnowski TJ (1995) A model of spike initiation in neocortical pyramidal neurons. *Neuron* 15, 1427-39.
3. Mainen ZF, Sejnowski TJ (1995) Reliability of spike timing in neocortical neurons. *Science* 268, 1503-6.
2. Destexhe A, Mainen ZF, Sejnowski TJ (1994) An efficient method for computing synaptic conductances based on a kinetic model of receptor binding. *Neural Computation* 6, 14-8.
1. Destexhe A, Mainen ZF, Sejnowski TJ (1994) Synthesis of models for excitable membranes, synaptic transmission, and neuromodulation using a common kinetic framework. *J Computational Neurosci.* 1, 195-230.

Reviews & Commentary

23. Wool LE, International Brain Laboratory (2020) Knowledge across networks: how to build a global neuroscience collaboration. *Curr. Opin. Neurobiol.* 65, 100-107.
22. Abbott LF, Angelaki DE, Carandini M, Churchland AK, Dan Y, Dayan P, Deneve S, Fiete I, Ganguli S, Harris KD, Häusser M, Hofer S, Latham PE, Mainen ZF, Mrosovsky T, Paninski L, Pillow JW, Pouget A, Svoboda K, Witten IB, Zador AM. (2017) An International Laboratory for Systems and Computational Neuroscience. *Neuron* 96(6):1213-1218.
21. Mainen ZF, Häusser M, Pouget A. (2016) A better way to crack the brain. *Nature* 539, 159-161
20. Gomez-Marín A, Mainen ZF. (2016) Expanding perspectives on cognition in humans, animals, and machines. *Curr. Opin. Neurobiol.* 37, 85–91.
19. Meyniel F, Sigman M, Mainen ZF. (2015) Confidence as bayesian probability: from neural origins to behavior. *Neuron* 88 (1), 78–92.
18. Murakami M, Mainen ZF. (2015) Preparing and selecting actions with neural populations: toward cortical circuit mechanisms. *Curr. Opin. Neurobiol.* 33 (C), 40–46.
17. Mainen ZF, Pouget A (2014) Put brain project back on course. *Nature* 511,534.
16. Gomez-Marín A, Paton JJ, Kampff AR, Costa RM, Mainen ZF. (2014) Big behavioral data: psychology, ethology and the foundations of neuroscience. *Nat. Neurosci.* (17), 1455–1462.
15. Arber S, Ehlers MD, Frégnac Y, Mainen ZF, Raymond J, Isaacs T (2013) BRAIN Initiative and Human Brain Project: Hopes and Reservations. *Cell* 155(2): 265-266.
14. Kepecs A, Mainen ZF (2012) A computational framework for the study of confidence in humans and animals. *Philos. Trans. R. Soc. Lond. B. Biol. Sci.* 367 (1594), 1322-37.
13. Vicente MI, Mainen ZF (2011) Convergence in the piriform cortex. *Neuron* 70 (1), 1-2.
12. Feierstein CE, Mainen ZF (2010) Listening to the crowd: neuronal ensembles rule. *Neuron* 66 (3), 334-6.
11. Dugué GP, Mainen ZF (2009) How serotonin gates olfactory information flow. *Nat. Neurosci.* 12, 673-5.
10. Mainen ZF, Kepecs A (2009) Neural representation of behavioral outcomes in the orbitofrontal cortex. *Curr Opin Neurobiol.* 19 , 84-91.
9. Vicente MI, Mainen ZF (2008) Towards an image of a memory trace. *Front Sys Neurosci.* Dec 08 (2), 131.
8. Mainen ZF (2007) The main olfactory bulb and innate behavior: different perspectives on an olfactory scene. *Nat Neurosci.* 10, 1511-2.
7. Kepecs A, Uchida N, Mainen ZF (2006) The sniff as a unit of olfactory processing. *Chem Senses* 31, 167-79.
6. Mainen ZF (2006) Behavioral analysis of olfactory coding and computation in rodents. *Curr Opin Neurobiol.* 16, 429-34.
5. Wilson R, Mainen ZF (2006) Early events in olfactory processing. *Annu Rev Neurosci.* 29, 163-201.

4. Uchida N, Kepecs A, Mainen ZF (2006) Seeing at a glance: Limits on temporal integration in perception. *Nature Rev Neurosci.* 7, 485-91.
3. Mainen ZF (2000) Thinking about feeling. *Nature Neurosci.* 3, 857.
2. Malinow R, Mainen ZF, Hayashi Y (2000) LTP mechanisms: from silence to four-lane traffic. *Curr Opin Neurobiol.* 10, 352-7.
1. Svoboda K, Mainen ZF (1999) Synaptic calcium: Intracellular stores spill their guts. *Neuron* 22, 427-30.

Book Chapters

11. Rigato, J., Murakami, M., Mainen, Z. (2015), "Spontaneous decisions and free will: Empirical results and philosophical considerations", in *Cold Spring Harbor Symposia on Quantitative Biology: Cognition*, Volume 79: 177-184, Cold Spring Harbor Laboratory Press.
10. Kepecs A, Mainen ZF (2014) A Computational Framework for the Study of Confidence Across Species. In: *The Cognitive Neuroscience of Metacognition* (pp. 115-145). Springer Berlin Heidelberg.
9. Mainen ZF, Abbott LF (2007) Functional plasticity at dendritic synapses. In: *Dendrites*, 2nd Ed. (Stuart G, Spruston N, Häusser M, eds). Oxford: Oxford University Press.
8. Mainen ZF (1999) Functional plasticity at dendritic synapses. In: *Dendrites* (Stuart G, Spruston N, Häusser M, eds). Oxford: Oxford University Press, pp 310-338.
7. Mainen ZF, and Sejnowski TJ (1998) Modeling active dendritic processes in pyramidal neurons. In: *Methods in Neuronal Modeling*, 2nd Ed. (Koch C, Segev I, eds) Cambridge, MA: MIT Press, pp 171-210.
6. Destexhe A, Mainen ZF, Sejnowski TJ (1998) Kinetic models of synaptic transmission. In: *Methods in Neuronal Modeling*, 2nd Ed. (Koch C, Segev I, eds), Cambridge, MA: MIT Press, pp 1-26.
5. Destexhe A, Mainen ZF, Sejnowski TJ (1995) Elementary kinetics provide efficient models of synaptic transmission and neuromodulation. In: *The Neurobiology of Computation* (Bower J, ed) Norwell, MA: Kluwer Academic Press, pp 9-14.
4. Destexhe A, Mainen ZF, Sejnowski TJ (1995) Synaptic currents, neuromodulation and kinetic models. In: *The Handbook of Brain Theory and Neural Networks* (Arbib M, ed) Cambridge, MA: MIT Press, pp 956-959.
3. Brown TH, Zador AM, Mainen ZF, Claiborne BJ (1992) Computational models of the principal classes of hippocampal neurons. In: *Single Neuron Computation* (McKenna T, Davis J, Zornetzer SF, eds) New York: Academic Press, pp 61-80.
2. Kairiss EW, Mainen ZF, Claiborne BJ, Brown TH (1992) Dendritic control of hebbian computations. In: *Analysis and Modeling of Neural Systems* (Eeckman FH, ed) Boston: Kluwer Academic Publishers.
1. Brown TH, Mainen ZF, Zador AM, Claiborne BJ (1991) Self-organization and non-linear processing in hippocampal neurons. In: *Advances in neural information processing systems 3* (Lippman RP, Moody JE, Touretzky DS, eds) San Mateo, CA: Morgan Kaufmann, pp 39-45.

Abstracts

2022

Cazettes F, Renart A, Mainen ZF. Facial movements and their neural correlates reveal latent decision variables in mice. Cosyne, Lisbon, Portugal.

Stensola H*, Stensola T*, Patwa M, DeWitt E, Mainen ZF. State prediction in the primary olfactory cortex, Cosyne (online).

Poo C. Cortical circuits for olfactory behavior. Simons Collaboration on Global Brain Annual Meeting, Simons Foundation, USA.

Poo C. Cortical circuits for olfactory behavior. UCLA Neuroscience Theme Search Symposium, Los Angeles, USA.

2021

Stensola T*, Stensola H*, Patwa M, DeWitt E, Mainen ZF. State predictions in the primary olfactory cortex, SfN 2021 (online).

Bonacchi N, Bruijns S, Chapuis GA, Churchland AK, Faulkner M, Lee H, Meijer GT, Paninski, Roth NA, Steinmetz M, Taheri S, West J, Winter O, ... International Brain Laboratory. Reproducibility of electrophysiological measurements during decision-making, SfN (online).

Poo C, Agarwal G, Bonacchi N, Mainen ZF. Spatial maps in piriform cortex during olfactory navigation, Spanish Society of Neuroscience Meeting, Spain.

Cazettes F, Murakami M, Renart A, Mainen ZF. Multiplexed accumulation of evidence in the mouse brain. Japan Neuroscience Society (online).

Poo C. Cortical circuits for olfactory behavior, Cornell NeuroAI Symposium, NY, USA.

Poo C. Cortical circuits for olfactory behavior, Allen Institute Annual Showcase, Seattle, USA.

Cazettes F, Murakami M, Renart A, Mainen ZF. Multiplexing decision computations in the mouse frontal cortex. Champalimaud Research Symposium: Dialogues on Neural and Machine Intelligence, Lisbon, Portugal.

Meijer GT, Laranjeira I, Findling C, International Brain Laboratory, Pouget A, Mainen ZF. Stimulating serotonergic neurons in the dorsal raphe nucleus during decision making results in stronger priors which are updated more rapidly. Champalimaud Research Symposium: Dialogues on Neural and Machine Intelligence, Lisbon, Portugal.

Poo C, Agarwal G, Bonacchi N, Mainen ZF. Spatial maps in piriform cortex during olfactory navigation, Champalimaud Research Symposium: Dialogues on Neural and Machine Intelligence, Lisbon, Portugal.

Stensola H*, Stensola T*, Patwa M, DeWitt E, Mainen ZF. State predictions in the primary olfactory cortex, Champalimaud Research Symposium: Dialogues on Neural and Machine Intelligence (online).

Cazettes F, Murakami Masayoshi, Renart A, Mainen ZF. A basis set for accumulation of action outcomes in the mouse frontal cortex. European Brain and Behavior Society (online).

Cazettes F, Murakami M, Renart A, Mainen ZF. Parallel inference in the mouse frontal cortex. Cosyne (online).

2020

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Poo C, Agarwal G, Bonacchi N, Mainen ZF. A map for odors and place in posterior piriform cortex, Annual Meeting of the Japan Neuroscience Society (online).

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1995

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Huguenard JR, Joerges J, Mainen ZF, Sejnowski TJ. Functional role of axon hillock and initial segment in cortical spike initiation. Soc Neurosci Abstr 20, 1527.

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1991

Mainen ZF, Claiborne BJ, Brown TH. A novel role for synaptic competition in the development of cortical lamination. Soc Neurosci Abstr 17, 759.

Mainen ZF, Claiborne BJ, Brown TH. Evolution of feature clusters in hippocampal dendrites. Proceedings of American Inst. Physics: Neural Networks for Computing, Snowbird, Utah, USA.

1990

Mainen ZF, Zador AM, Claiborne BJ, Brown TH. Hebbian synapses induce feature mosaics in hippocampal dendrites. Soc Neurosci Abstr 16, 492.

Technical reports

Bell AJ, Mainen ZF, Tsodyks M, Sejnowski TJ (1995) "Balancing" of conductances may explain irregular cortical spiking. Institute for Neural Computation, University of California, San Diego, Technical report #9502.

Patents

Min SL, Tanno CL, Mainen ZF, Softky WR (2003) System and method for context-based document retrieval. United States Patent 6,633,868.

Zittrain J, Min SL, Mainen ZF, Lambert MW, Goetsch D (2006) State adaptation devices and methods for wireless communications. United States Patent 6,998,984.

Thesis Supervision

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2000-2007	Sachin Ranade, PhD, Stony Brook University, USA.
2000-2007	Hatim Zariwala, PhD, Stony Brook University, USA.
2001-2007	Claudia Feierstein, PhD, Watson School of Biological Sciences, Cold Spring Harbor Laboratory, New York, USA.

Lab Members

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2011-2017	Paul Bush – Postdoctoral fellow, Champalimaud Centre for the Unknown, Lisbon, Portugal.
2010-2018	Eric DeWitt – Postdoctoral fellow, Champalimaud Centre for the Unknown, Lisbon, Portugal.
2006-2018	Masayoshi Murakami – Postdoctoral fellow, Cold Spring Harbor Laboratory, New York, USA/ Champalimaud Centre for the Unknown, Lisbon, Portugal.
2015–2017	Joana Rigato – Postdoctoral fellow, Champalimaud Centre for the Unknown, Lisbon, Portugal.
2014-2017	Ana Nunes – Lab Manager, Champalimaud Centre for the Unknown, Lisbon, Portugal.
2014-2016	Eva Kobak – Postdoctoral fellow, Champalimaud Centre for the Unknown, Lisbon, Portugal.
2011-2017	Rita Venturini – Invited Scientist, Champalimaud Centre for the Unknown, Lisbon, Portugal.
2011-2014	Enrica Audero – Postdoctoral Fellow/Project Manager, Champalimaud Centre for the Unknown, Lisbon, Portugal.
2009-2013	Magor Lorincz – Postdoctoral fellow, Champalimaud Centre for the Unknown, Lisbon, Portugal.
2009-2012	Hope Johnson – Postdoctoral fellow, Champalimaud Centre for the Unknown, Lisbon, Portugal.
2007-2012	Guillaume Dugué – Postdoctoral fellow, Champalimaud Centre for the Unknown, Lisbon, Portugal.
2006-2008	Ninglong Xu – Postdoctoral fellow, Cold Spring Harbor Laboratory, New York, USA
2006-2008	Matthew Smear – Postdoctoral fellow, Cold Spring Harbor Laboratory, New York, USA
2006-2007	Joshua Sanders – Research Assistant, Cold Spring Harbor Laboratory, New York, USA
2005-2009	Gidon Felsen – Postdoctoral fellow, Cold Spring Harbor Laboratory, New York, USA
2003-2008	Adam Kepecs – Postdoctoral fellow, Cold Spring Harbor Laboratory, New York, USA

2003-2005	Dara L. Sosulski – Research Assistant, Cold Spring Harbor Laboratory, New York, USA
2002	Nerege Okraien - Summer Student, Cold Spring Harbor Laboratory, New York, USA
2001-2006	Michael Quirk – Postdoctoral fellow, Cold Spring Harbor Laboratory, New York, USA
2001-2002	Gautam Agarwal – Summer Student, Cold Spring Harbor Laboratory, New York, USA
2000-2005	Hirac Gurden – Postdoctoral fellow, Cold Spring Harbor Laboratory, New York, USA
2000-2006	Naoshige Uchida – Postdoctoral fellow, Cold Spring Harbor Laboratory, New York, USA
2000-2004	Veronica Egger – Postdoctoral fellow, Cold Spring Harbor Laboratory, New York, USA
2000-2002	Natalia Caporale – Research Assistant, Cold Spring Harbor Laboratory, New York, USA
1999-2001	Steve Macknik – Postdoctoral fellow, Cold Spring Harbor Laboratory, New York, USA
1999-2001	Robert Gaspar – Technician
1999-2000	Eric Sullivan - Student

Teaching

2016	Instructor Neuroengineering course Instituto Superior Técnico, Universidade de Lisboa, Portugal
2014-2018	Founder and Co-Organizer CAJAL Advanced Neuroscience Training Programme (FENS-IBRO) Champalimaud Centre for the Unknown, Lisbon, Portugal
2010-2018	Founder and Co-Director Computational and Cognitive Neuroscience Summer School (CCNS) Cold Spring Harbor Asia, Suzhou & Beijing, China
2012-present 2007-2011	Instructor Founder, Director & Instructor International Neuroscience Doctoral Programme Champalimaud Centre for the Unknown, Lisbon, Portugal
2002-2003	Lead instructor, Systems Neuroscience
2002-2004	Lecturer, Scientific Reasoning and Logic
2001-2006	Instructor, Systems Neuroscience Watson School of Biological Sciences Cold Spring Harbor Laboratory, Cold Spring Harbor, NY, USA

Meeting Organization

2017	Co-Organizer, <i>MDMA and psilocybin: Bridging science and clinical application</i> . Cascais, Portugal
2015	Co-Organizer, <i>Subjective Confidence: psychology, physiology, theory</i> , Fondation Les Treilles, Tourtour, France
2008	Co-Organizer, <i>Neural Circuits and Decision-Making in Rodents</i> , Janelia Farm, Ashburn, VA, USA
2007	Co-Organizer, <i>Theoretical and Experimental Perspectives on Serotonin Function</i> , Ericeira, Portugal
2005-present	Executive Board, <i>Computational and Systems Neuroscience</i> (Cosyne), Salt Lake City, UT, USA
2006	General Chair, <i>Computational and Systems Neuroscience</i> (Cosyne), Salt Lake City, UT, USA
2005	Program Chair, <i>Computational and Systems Neuroscience</i> (Cosyne), Salt Lake City, UT, USA
2005	Co-Organizer, <i>Neural Information and Coding International</i> (NICI), Karnataka, India
2001	Co-Organizer, <i>Neural Information and Coding Europe</i> (NICE), Les Houches, France
2000	Co-Organizer, <i>Banbury Workshop on Neural networks and Cognition</i> , Cold Spring Harbor, NY, USA

Invited Seminars

2020

- Role of the dorsal raphe serotonergic system on decision-making in a foraging context. FENS Virtual Forum

2019

- “Thoughts on the future of humans and machines”, Emotion and Morality: (Un)conscious Journey to Being VI, NOVA University, Lisbon, Portugal
- “Serotonin and the regulation of adaptive behavior”, Ascona Meeting on Neuronal Circuits, Ascona, Switzerland
- “Thoughts on the future of humans and machines”, Brain: wider than the sky, Calouste Gulbenkian Foundation, Lisbon, Portugal

2018

- 83rd Cold Spring Harbor Symposium: “Brains & Behavior: Order & Disorder in the Nervous System”, Cold Spring Harbor, New York, USA
- “Canonical Computational in Brains and Machines” Symposium, NY University, New York City, USA
- SfN 2018 Special Lecture, San Diego, CA, USA

2017

- “Chemical Neuromodulation: Neurobiological, Neurocomputational, Behavioural and Clinical Aspects”, FENS SfN Summer School, Bertinoro, Italy
- “Sussex Neuroscience Day”, University of Sussex, UK

2016

- Centre for Theoretical Neuroscience, Columbia University, New York City, USA
- “Complexity of Neural Computation and Cognition” workshop, Janelia Farm Research Campus, Ashburn, VA, USA
- FENS Forum, Copenhagen, Denmark
- Gatsby Computational Neuroscience Unit, University College London, London, UK
- “Uncertainty and Action” workshop, Kavli Center at Chicheley Hall, The Royal Society, Chicheley, UK
- “Sensory logic of the Gastronomic Brain” workshop, San Sebastian, Spain
- CEDOC, NOVA Medical School, Lisbon, Portugal

2015

- Canonical Neural Computations, NYU, Florence, Italy
- “International Brain Projects”, Cold Spring Harbor Asia, Suzhou, China
- Charité - Universitätsmedizin Berlin, Germany
- The Gatsby Computational Neuroscience Unit, University College London, UK
- ICONE - International Conference on Neuroethics, Lisbon, Portugal
- CoSyNe workshops, Snowbird, USA
- FENS-Hertie Winter School "The neuroscience of decision making", Obergurgl, Austria

2014

- SISSA, Trieste, Italy
- “The aims of brain research: scientific and philosophical perspective”, The 28th Annual International Workshop on the History and Philosophy of Science, Van Leer Jerusalem Institute, Israel
- 1st joint Postdoc meeting IMM, IGC, CEDOC, Sintra, Portugal
- Boom Festival, Idanha-a-Nova, Portugal
- “Mapping the Unknown”, Ar Event, Champalimaud Centre for the Unknown, Lisbon, Portugal
- 79th Cold Spring Harbor Symposium on Quantitative Biology “Cognition”, Cold Spring Harbor, New York, USA
- “Controlling Neurons, Circuits, Behavior”, FENS Spring Brain Conference, Rungstedgaard, Copenhagen, Denmark
- 16th International Neuroscience Winter Conference, Sölden, Austria
- CoSyNe, Salt Lake City, USA
- CoSyNe workshops, Snowbird, USA
-

2013

- Bernstein Center for Computational Neuroscience, Tübingen, Germany
- Institute of Neuroinformatics, University of Zurich, Zurich, Switzerland
- *3rd Latin American School for Education, Cognitive and Neural Sciences*, Comandatuba, Brazil
- *1st European Neuroscience Conference by Doctoral Students*, Bordeaux, France
- *Temporal Dynamics in Learning: Networks and Neural Data*, HHMI Janelia Farm Research Campus, USA
- *Noise in decision-making*, Barcelona, Spain
- *A Multilevel Approach to Distributed Cognition*, Cascais, Portugal
- Yale University, New Haven, CT, USA
- Institute of Neuroscience, Shanghai, China

2012

- CoSyNe, Salt Lake City, USA
- CoSyNe workshops, Snowbird, USA
- Cold Spring Harbor Laboratory, Cold Spring Harbor, NY, USA
- Stanford University, USA
- NERF Neurotechnology Symposium, Leuven, Belgium
- Brain dynamics and decision making, Ascona, Switzerland

- Canonical Neural Computation, Florence, Italy

2011

- Columbia University, New York, USA
- Société des Neurosciences 10e Colloque, Marseille, France
- FENS-IBRO-SfN School, Bertinoro, Italy
- GABBA Annual Symposium, Porto, Portugal
- *Genes, circuits, behaviour Symposium*, RIKEN Brain Science Institute, Tokyo, Japan

2010

- Brain Circuits Workshop, Ein Gedi, Israel
- Weizmann Institute, Rehovot, Israel
- CoSyNe Workshops, Snowbird, Utah, USA
- Tamagawa-Caltech Lecture Course on Decision-making, Tamagawa University, Tokyo, Japan
- 1st São Paulo School of Translational Sciences, AC Camargo Hospital, São Paulo, Brazil
- Form and Function of the Olfactory System Workshop, Janelia Farm Research Campus, Ashburn, VA, USA
- Oxford University, Oxford, UK
- Photons and Neurons, University of Rochester, Rochester, NY, USA
- Computational and Cognitive Neuroscience Course, CSH Asia, Suzhou, China
- EMBO|EMBL Symposium: Structure and Function of Neural Circuits, Heidelberg, Germany
- Instituto de Neurociencias, Alicante, Spain
- Adrian Seminar, University of Cambridge, Cambridge, UK

2009

- CNRS Universit. Claude Bernard, Lyon, France
- Pan-Iberian Systems Neuroscience Meeting: Communicating in Neural Circuits, Lisbon, Portugal
- EMBO Conference Series: The Assembly and Function of Neuronal Circuits, Ascona, Switzerland
- Lemanic European Neuroscience Institute Network Meeting, Geneva, Switzerland
- Interdisziplinäres Zentrum fuer Neurowissenschaften, University of Heidelberg, Heidelberg, Germany
- Computational Neuroscience Meeting: Quantitative Models of Behaviour Workshop, Berlin, Germany
- Okinawa Computational Neuroscience Course, Okinawa, Japan
- Simpósio Luso Brasileiro-Neurociência, Braga, Portugal
- Friedrich Miescher Institute for Biomedical Research, Basel, Switzerland
- Simpósio “Psicologia e Neurociências” at Instituto Superior de Psicologia Aplicada, Lisbon, Portugal
- Frontiers in Biology Symposium, IGC, Oeiras, Portugal
- Keystone Symposium on Chemical Senses: Receptors and Circuits, Tahoe City, California, USA
- Department of Neurobiology, Duke University, Durham, NC, USA
- Instituto de Biologia Molecular e Celular (IBMC), Porto, Portugal
- FENS-Hertie Winter School: Structure and Function of Neural Circuits, Obergurgl, Austria

2008

- Fórum Internacional de Investigadores Portugueses, IMM, Lisbon, Portugal
- Department of Neuroscience, Stanford University, Stanford, CA, USA
- IMP, Vienna, Austria
- IDIBAPS, Barcelona, Spain, USA
- Ecole Normal Supérieur, Paris, France
- Instituto Superior Psicologia Aplicada, Lisbon, Portugal, USA
- Neural circuits and decision making in rodents, Janelia Farm, Ashburn, VA, USA

2007

- Department of Neurobiology, Yale University, New Haven, CT, USA
- Center for Neural Science and Department of Economics, New York University, New York, NY, USA

- Symposium on 5-HT, Society for Neuroscience Meeting, San Diego, CA, USA
- Department of Neuroscience, University of California San Diego, La Jolla, CA, USA

2006

- Department of Molecular and Cell Biology, University of California, Berkeley, CA, USA
- Department of Physiology, Columbia Univ., New York, NY, USA
- Department of Neuroscience, Barrow Inst., Phoenix, AZ, USA

2005

- Gordon Conference, Salve Regina University, Newport, RI, USA
- Computation and Neural Systems, Caltech, Pasadena, CA, USA

2004

- Neuroscience Departmental Retreat, Stanford University, Stanford, CA, USA
- Fondation des Treilles, Tourtour, France
- Methods in Computational Neuroscience, MBL, Woods Hole, MA, USA
- Banbury Center, Cold Spring Harbor, NY, USA
- Monell Chemical Senses Center, Philadelphia, PA, USA

2003

- Olfactory Coding Tutorial, Neural Information Processing Systems, Vancouver, BC, Canada
- Computation and Neural Systems Meeting, Alicante, Spain
- Redwood Neuroscience Institute, Redwood, CA, USA
- National Center for Biological Sciences, Bangalore, India

2002

- Gatsby Computational Neuroscience Unit, London, England
- Berlin Free University, Berlin, Germany